Conditional borrower cash flows and liquidation close-out

Niko Rokni Lamouki | Conditional accounting note

State and counterfactual. Original draw P is USD-equivalent DAI; a hypothetical local-currency debt starts at PE_0, where E is local currency per USD. The effective annual local borrowing rate r gives $D(t) = P(E_0/E_t)(1+r)^t$. The comparator is the same borrower retaining the original collateral without borrowing. Draw proceeds are assumed held at par; profits, reinvestment, taxes and actual local-token trades are not observed.

Opening costs. At $t = 0$, an externally funded fee fP, opening conversion friction sP, and opening gas $g/2$ are paid. The fee is not financed. Base assumptions are $r = 20\%$, $f = 0.5\%$, $s = 0.3\%$ each conversion, and $g = \$40$ split equally between opening and voluntary closing. They are illustrative, not measured prices.

Mutually exclusive settlement routes

$$\begin{aligned}
&\text{Voluntary full repayment at } h: && B^{\text{repay}}(h) = P - fP - sP - g/2 - D(h) - sD(h) - g/2,\\
&\text{Sale at liquidation time } \tau: && S = (1-a)V_\tau, \quad R = \min(S, D(\tau)), \quad W = D(\tau) - R,\\
& && \Pi = \min\{\lambda D(\tau), [S - D(\tau)]_+\},\\
& && H = [S - D(\tau)]_+ - \Pi, \quad B^{\text{liq}}(\tau) = P - fP - sP - g/2 - V_\tau + H.
\end{aligned}$$

V_τ is the owned collateral mark, a an auction haircut, R debt recovery, W issuer/lender asset impairment, Π a penalty capped by surplus, and H collateral proceeds returned to the borrower. The liquidated branch has *no further borrower repayment* at h, no voluntary exit swap, and no voluntary closing gas. The shortfall is assigned to claimholders under a stated nonrecourse assumption; a recourse contract would need a different ledger.

Numerical reconciliation, excluding all opening costs and haircut.

At sale: $P = 100$, $D = 100$	V	R	Π	H	W	B^{liq}
Collateral surplus; $\lambda = 10\%$	180	100	10	70	0	-10
Collateral shortfall; $\lambda = 13\%$	60	60	0	0	40	40

For every sale, $R + W = D$ and $R + \Pi + H = S$. Borrower collateral opportunity cost is $V - H$, never $V - H$ *plus* a separate repayment. These identities are checked in the executable tests.

Empirical identification boundary. The data supply official monthly FX and daily published reference marks for selected spells, but not an intraday executable quote, within-horizon collateral path, actual τ, auction price or probability weights. The reported collateral result is solely a terminal ratio $c_0P(1-q)/D(1)$ below a stipulated threshold. It is not an executed-liquidation probability or expected borrower outcome.